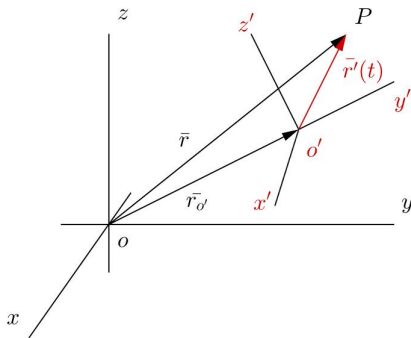


Equation of Motion in a Non-Inertial FoR



xyz — inertial FoR

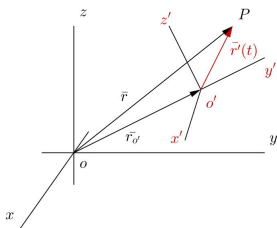
$x'y'z'$ — another FoR that moves arbitrarily w.r.t. xyz (may be accelerating, rotating with varying angular velocity, . . .)

Previously, we have considered a situation when $x'y'z'$ was moving with a constant velocity w.r.t xyz (then $x'y'z'$ is inertial as well). Now, it moves arbitrarily, hence is not inertial.

Goal: Derive a (kinematic) relationship between accelerations of a particle P in both frames of reference.

Relationship between position vectors in both FoR

$$\vec{r} = \vec{r}_{O'} + \vec{r}'$$



where

- $\vec{r}$ — position of the particle in xyz (we use fixed unit vectors $\hat{n}_x, \hat{n}_y, \hat{n}_z$ here)
- $\vec{r}_{O'}$ — position of the origin of $x'y'z'$ as seen from xyz;
- $\vec{r}'$ — position of the particle in $x'y'z'$ (here, we have to use $\hat{n}_{x'}, \hat{n}_{y'}, \hat{n}_{z'}$, that are not fixed; in particular may rotate)

Complication: Motion of $x'y'z'$ is arbitrary, will need to take into account that $\hat{n}_{x'}, \hat{n}_{y'}, \hat{n}_{z'}$ are not fixed (i.e. will need to know how to calculate their derivatives w.r.t. time).

Step 1. Relationship Between Velocities

Useful notation (**Einstein's Summation Convention**)

$$r_\alpha \hat{n}_\alpha \quad \text{denotes} \quad \vec{r} = \sum_{\alpha=x,y,z} r_\alpha \hat{n}_\alpha = x\hat{n}_x + y\hat{n}_y + z\hat{n}_z \quad (\text{skip the "}\Sigma\text{"})$$

Using this convention, we have

$$\vec{r}(t) = r_\alpha \hat{n}_\alpha, \quad \vec{r}'(t) = r_{\alpha'} \hat{n}_{\alpha'}.$$

Differentiate both sides of $\vec{r} = \vec{r}_{O'} + \vec{r}'$ w.r.t. time to find

$$\frac{d\vec{r}}{dt} = \vec{v} = \frac{d\vec{r}_{O'}(t)}{dt} + \frac{d\vec{r}'(t)}{dt} = \vec{v}'_O + \frac{d\vec{r}'(t)}{dt}$$

with

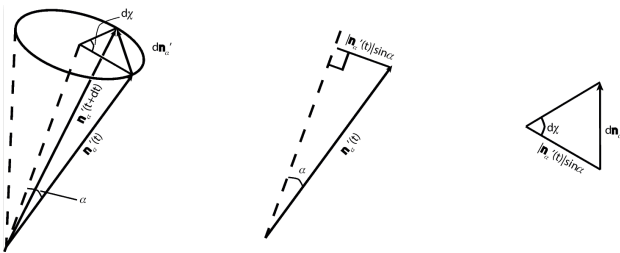
$$\frac{d\vec{r}'(t)}{dt} = \frac{d}{dt}(r_{\alpha'} \hat{n}_{\alpha'}) = \dot{r}_{\alpha'} \hat{n}_{\alpha'} + r_{\alpha'} \dot{\hat{n}}_{\alpha'}$$

Note that $\dot{r}_{\alpha'} \hat{n}_{\alpha'} = v_{\alpha'} \hat{n}_{\alpha'} = \vec{v}'$.

But what is $\dot{\hat{n}}_{\alpha'}$ equal to?

Differentiation of Unit Vectors in the Non-Inertial FoR

Observation. The length of $\hat{n}_{\alpha'}$ is fixed, therefore only its direction can change. The derivative $\dot{\hat{n}}_{\alpha'}$ is associated with rotations of the non-inertial FoR.



Let α be the angle between the instantaneous axis of rotation and $\hat{n}_{\alpha'}$. $d\chi$ is then the corresponding angle that is swept by the projection of $\hat{n}_{\alpha'}$ (radius of the circle in the left part of the figure) over the infinitesimal time dt . Therefore, the magnitude of the infinitesimal change $d\hat{n}_{\alpha'}$ is

$$|d\hat{n}_{\alpha'}| = d\chi |\hat{n}_{\alpha'}| \sin \alpha.$$

The rotation angle $d\chi$ can be defined as a vector $d\overline{\chi}$, with the direction along the instantaneous axis of rotation (use the right-hand rule). Then

$$d\hat{n}_{\alpha'} = d\overline{\chi} \times \hat{n}_{\alpha'}$$

and, consequently,

$$\boxed{\frac{d\hat{n}_{\alpha'}}{dt} = \frac{d\overline{\chi}}{dt} \times \hat{n}_{\alpha'},}$$

where $\overline{\omega} = \frac{d\overline{\chi}}{dt}$ is the **instantaneous angular velocity**.

The differentiation rule for the unit vectors of the non-inertial FoR with respect to time can eventually be written down as

$$\boxed{\dot{\hat{n}}_{\alpha'} = \overline{\omega} \times \hat{n}_{\alpha'}.$$

Step 1. Relationship Between Velocities

Returning to the derivation of the relationship between velocities and using the differentiation rule for unit vectors, we have

$$\begin{aligned}\frac{d\vec{r}'(t)}{dt} &= \vec{v}' + r_{\alpha'} \dot{\hat{n}}_{\alpha'} \\ &= \vec{v}' + r_{\alpha'} (\bar{\omega} \times \hat{n}_{\alpha'}) \\ &= \vec{v}' + \bar{\omega} \times r_{\alpha'} \hat{n}_{\alpha'} \\ &= \vec{v}' + \bar{\omega} \times \vec{r}'.\end{aligned}$$

Plugging back into the relationship between velocities we obtain

$$\boxed{\vec{v} = \vec{v}_{O'} + \vec{v}' + (\bar{\omega} \times \vec{r}')}.$$

Comment. The three terms on the rhs correspond to: the translational motion of the primed FoR, relative velocity of the particle w.r.t. that FoR, and to rotations of the primed FoR.

Step 2. Relationship Between Accelerations

Differentiating both sides of the relationship between velocities

$\bar{\mathbf{v}} = \bar{\mathbf{v}}_{O'} + \bar{\mathbf{v}}' + (\bar{\boldsymbol{\omega}} \times \bar{\mathbf{r}}')$ term by term, we obtain

$$\bar{\mathbf{a}} = \frac{d\bar{\mathbf{v}}}{dt} = \frac{d\bar{\mathbf{v}}_{O'}}{dt} + \boxed{\frac{d\bar{\mathbf{v}}'}{dt}} + \frac{d\bar{\boldsymbol{\omega}}}{dt} \times \bar{\mathbf{r}}' + \boxed{\bar{\boldsymbol{\omega}} \times \frac{d\bar{\mathbf{r}}'}{dt}}.$$

In evaluating the former boxed term, we use the differentiation rule for unit vectors once again

$$\begin{aligned} \frac{d\bar{\mathbf{v}}'}{dt} &= \frac{d}{dt}(v_{\alpha'} \hat{\mathbf{n}}_{\alpha'}) = \dot{v}_{\alpha'} \hat{\mathbf{n}}_{\alpha'} + v_{\alpha'} \dot{\hat{\mathbf{n}}}_{\alpha'} = \bar{\mathbf{a}}' + v_{\alpha'} \bar{\boldsymbol{\omega}} \times \hat{\mathbf{n}}_{\alpha'} \\ &= \bar{\mathbf{a}}' + \bar{\boldsymbol{\omega}} \times v_{\alpha'} \hat{\mathbf{n}}_{\alpha'} = \boxed{\bar{\mathbf{a}}' + \bar{\boldsymbol{\omega}} \times \bar{\mathbf{v}}'}. \end{aligned}$$

Similarly, for the latter one,

$$\begin{aligned} \bar{\boldsymbol{\omega}} \times \frac{d\bar{\mathbf{r}}'}{dt} &= \bar{\boldsymbol{\omega}} \times \frac{d}{dt}(r_{\alpha'} \hat{\mathbf{n}}_{\alpha'}) = \bar{\boldsymbol{\omega}} \times (\dot{r}_{\alpha'} \hat{\mathbf{n}}_{\alpha'} + r_{\alpha'} \dot{\hat{\mathbf{n}}}_{\alpha'}) \\ &= \boxed{\bar{\boldsymbol{\omega}} \times \bar{\mathbf{v}}' + \bar{\boldsymbol{\omega}} \times (\bar{\boldsymbol{\omega}} \times \bar{\mathbf{r}}')}. \end{aligned}$$

Combining results from the previous slide we finally obtain the relation between accelerations

$$\bar{a} = \bar{a}_{O'} + \bar{a}' + 2\bar{\omega} \times \bar{v}' + \frac{d\bar{\omega}}{dt} \times \bar{r}' + \bar{\omega} \times (\bar{\omega} \times \bar{r}').$$

In order to understand its physical interpretation, we multiply both sides by m and rearrange the terms

$$\begin{aligned} m\bar{a}' &= m\bar{a} - m\bar{a}_{O'} - m\frac{d\bar{\omega}}{dt} \times \bar{r}' - 2m(\bar{\omega} \times \bar{v}') - m\bar{\omega} \times (\bar{\omega} \times \bar{r}') \\ &= \underbrace{\bar{F}}_{\text{real force}} - \underbrace{m\bar{a}_{O'} - m\frac{d\bar{\omega}}{dt} \times \bar{r}' - 2m(\bar{\omega} \times \bar{v}') - m\bar{\omega} \times (\bar{\omega} \times \bar{r}')}_{\text{pseudo-forces (also called fictitious forces or forces of inertia);}} \end{aligned}$$

real force
(of material origin)

pseudo-forces (also called *fictitious forces* or forces of inertia); these are kinematic corrections (which have units of [N]) that **must** be included due to the fact that we describe dynamics in a non-inertial FoR (*force* = *mass* × *acceleration* is valid only in inertial FoRs). These "forces" **never** appear in inertial FoRs!